

Chapter – 6: Pressure, Winds, Storms, and Cyclones

- Obs. – fallen leaves – ground – swirl – sweep away – tree – sway – bend – strong winds
- Ever wonder – such things – happen? – wind – applies force?
- Recall – similar effects – doors slamming – windows rattling – clothes fluttering
- Force – wind – wind pressure – cause effects
- This chapter – **explore** relations – force, pressure – understand causes – powerful natural events – thunderstorms, cyclones

Pressure

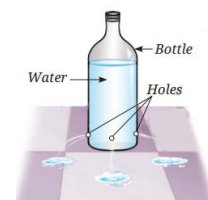
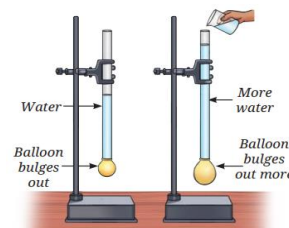
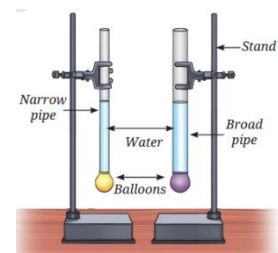
- Megha, Pawan – picnic – walk – picnic spot – identical items – bags
- On the way – Pawan – uncomfortable – adjust bag
- Megha – any problem – bag
- Pawan – shoulders hurt
- Megha – both bags – equally heavy – your hurts – mine doesn't
- Pawan – maybe – diff. straps – my bag – narrow strap – your bag – broad strap
- Shape, size – strap – matters?
- Carry bag – feel weight – gravity – shoulders
- Wt. of bag – **narrow straps** – **smaller area** – shoulders – BUT – wt. of bag – **broad strap** – spread out – **larger area** – shoulders
- This reason – more comfortable – carry bag – broad straps – instead – narrow straps – both bags – same wt.
- Area – force act – involved – define qty. – **pressure = force per unit area**
- Pressure = $\frac{\text{Force}}{\text{Area}}$
 - This stage – consider forces – perpendicular to surface
- Broad straps – reduce pressure – compare to – narrow straps – more comfortable – carry bag – broad straps
- People – carry loads – pots, vegetable baskets – heads – often – round piece – under loads
- Both cases – objective – reduce pressure – increase area – wt. acts
- Pressure – force per unit area – **SI Unit** – **force** – **newton** – that of **area** – **metre²**
- **SI Unit** – **pressure** – **newton / metre² (N / m²)** – also called – **pascal (pa)**
- Force – 100 N – cardboard area – 2 m² – pressure –
 - Pressure = $\frac{\text{force}}{\text{area}} = \frac{100 \text{ N}}{2 \text{ m}^2} = 50 \text{ N / m}^2$
- Many situations – daily life – pressure – imp. role
- **Conduct** – act. – below table – **record** obs. – **explain** – pressure – influence – action – each act.



Activity	Modes of action	Easy / difficult ? give reasons
Driving an iron nail	1. Head of nail 2. Pointed end of nail	1. Easy – less area contact wood 2. Difficult – more area contact wood
Cutting apple with knife	1. Use sharp edge 2. Use blunt edge	1. Easy – less area – more pressure

		2. Difficult – more area – less pressure
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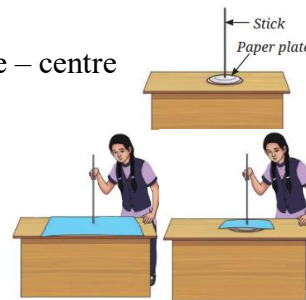
- Conclude – area – force applied – smaller – result pressure – higher – easier – certain tasks
- Easier – drive nail – pointed end – easier – cut apple – sharp edge – knife
- Overhead water tank – locality – rooftops – why – height?
- Activity –
 - Take – 2 transparent glass, plastic pipes – same length – diff. diameters
 - Take – 2 rubber balloons – attach 1 – each pipe
 - Clamp pipes – stand – fig.
 - Fill both pipes – water – same level – halfway
 - **Observe** – balloons – bulge – same extent?
- **Infer** – this act. – obs. – 2 balloons – bulge – same extent
- Diff. diameters – wt. of water – 2 pipes – diff. – BUT – bulge – both balloons – same
- Meaning – wt. of water – pipes – not responsible – bulge
- Pressure – water column – responsible – bulge – equal heights – equal bulge – EVEN – diff. diameters
- Pour water – any pipe – obs. bulge – balloon – repeat process – few times – more water each time – note bulge
- Obs. rel. – amt. of bulge – rubber balloon – AND – height of water column – pipe
- Obs. – bulge increases – height of water column increases
- Height – water column – increase – pressure – pipe bottom – increase – cause – balloon – bulge more
- Pressure of liquid – vessel – depend – height of column – reason – overhead tanks – height – pressure – taps – increase – result – good stream – water
- Activity –
 - Take – used plastic bottle – remove cap – make 4 holes – bottom – side – needle, nail – make sure – holes – same height from bottom – fig.
 - Seal holes – tape – fill bottle – water
 - Remove tape – all holes – same time – obs.
- Obs. – water flow out – holes – bottle side
- **Indicate** – water exert pressure – sides – container
- Conclude – liquid – pressure – bottom, sides – container – all dir.
- Broken pipe – water spurts – fountain – reason – pressure by water – walls of liquid
- **Interesting fact** –
 - Base of dam – much broader – top – much thinner
 - Broad base – support structure – BUT also – handle – horizontal water pressure
 - Water in dams – horizontal pressure – side walls – dam – vertical pressure – height of water level
 - Horizontal pressure – very large – bottom – withstand pressure – dam bottom – broader



Pressure Exerted by Air

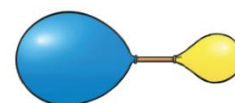
- Air – around us – envelope of air – surround earth – **atmosphere**
- Atmospheric air – nitrogen, oxygen, carbon dioxide, other gases – small qty.
- Atm. – many kms. – above surface
- Activity –

- Take – paper plate – invert – attach stick – fig. – place – plain surface
- Take – 2 sheets – chart paper – 70 x 56 cm – fold – 1 sheet – make hole – centre – big enough – stick comes out
- Place – folded sheet – top – inverted paper – fig.
- Try – lift plate – use stick – obs. – effort – lift it
- Place – 2nd unfolded sheet – make hole – centre – stick – pass through – cover plate – unfolded paper
- Lift plate – use stick – obs. – effort – lift it
- Obs. – more effort – unfolded paper
- Unfolded paper – covering area – increase – effort – lift it – increase
- Notice – wt. – covering sheet – not changed
- Infer – air exert force – covering sheet – difficult – lift plate
- Air – exert force – paper plate – increase – covering area – increase
- Pressure = force per unit area – conclude – air exert pressure
- Pressure – air – around us – **atmospheric pressure**
- Experience – blow air – balloon – inflated
- Air – exert pressure – balloon walls – all dir.
- Leave balloon – mouth open – air escape – why?
- Activity –
 - Take – good quality rubber sucker – press firmly – flat surface
 - Realise – stick – surface
 - Try – pull off – difficult?
- Press sucker – most air – b/w cup, surface – pushed out – air pressure – reduced
- Sucker – stick – surface – air pressure – surrounding – higher – air pressure – inside sucker
- Pull sucker – applied force – strong enough – overcome pressure diff. – outside, inside
- Force – atm. air column – area – 15 x 15 cm – equal – gravity force – obj. – mass – 225 kg (2250 N)
- Reason – we – not crushed – this wt. – inside pressure – our body – equal – atm. pressure
- Inside pressure – cause – movement – fluids, gases – tissues, organs
- **A Step Further**
 - SI unit – pressure – N / m² – pascal (Pa)
 - Practical unit – air pressure – millibar (mb) – equal – 100 Pa
 - Air pressure – also expressed – hectopascal (hPa) – equal – 100 Pa



Formation of Wind

- Notice – some days – wind – strong – other days – calm – sometimes – so strong – damage – life, prop.
- Inflated balloon – mouth open – air escapes
- Recall – puncture – bicycle tube – air escapes
- Both cases – air movement – high pressure to low pressure
- Activity –
 - Take – drinking straw, 2 similar balloons – thin rubber
 - Insert 1 end – straw – 1 balloon – secure – rubber band, thread
 - Inflate 2nd balloon – hold mouth – fingers – no air escape
 - Insert free end – straw – neck – inflated balloon – secure – rubber band, thread
 - Make sure – no air escape – now – straw – 1 end – inflated balloon – other end – uninflated balloon – fig.
 - **Predict** – what happens – balloons – obs. – what happens – same as predicted? – note down



- Reason – change in size – balloons? – Air pressure – inflated balloon – higher – uninflated balloon – result – air moves – inflated to uninflated balloon
- Sometime later – both balloons – same size – air flow – stopped
- Air flow – continue – difference – air pressure – stops – air pressure – same
- Conclude – air movement – higher pressure to lower pressure
- Relate conclusion – dir. – sea, land breeze – grade 7 – curiosity
- Land heats faster – day time – air over land – warmer, lighter – rise – area – low pressure
- Air – high pressure over sea – flows into land – result – sea breeze
- Night time – water – warmer – low pressure area – sea – result – wind – land to sea – land breeze
- Measure sp. – escaping air – find – higher pressure diff. – higher sp.

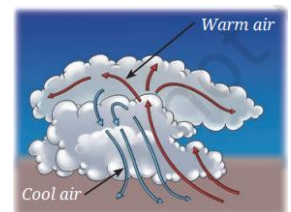
High-Speed Winds Result in Lowering of Air Pressure

- Activity –
 - Take – 2 balloons – same size
 - Inflate balloons – tie strings
 - Hang balloons – stick – leave gap – 6-10 cm
 - Blow air – narrow gap – obs.
 - Blow harder – obs.
- Blow b/w balloons – obs. – balloons – move closer
- Reason – blow air b/w balloons – create – low pressure area
- High air pressure – surround balloons – push together
- Obs. – hard blow – higher sp. – coming closer
- Infer – high sp. winds – accompanied – reduced air pressure
- High sp. winds – over houses – low pressure area – air pressure – above roof – lower than below
- Pressure diff. – large – roof – weak – blown away
- Much safer – doors, windows – open – high sp. winds – reduce pressure diff. – roofs kept safe
- High sp. winds – storms – accompanied – thunder, lightening

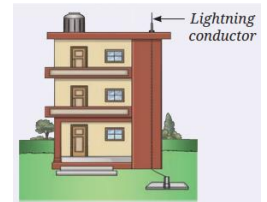


Storms, Thunderstorms, and Lightening

- Land – heats up – warm, moist air – lighter – rises – low pressure area
- Cool air – surrounding high pressure area – flows in
- This air – heats up – rise – result – continuous cycle – wind circulation
- Rising air – expand – cools – moisture condense – water droplets – form clouds
- Droplets – merge – heavier drops – come down – rain, hail, snow
- Strong winds with rain – **storm** – hot, humid, tropical regions – India – storms – more frequent
- Certain conditions – warm air – rise – great heights – low temp. – converts – droplets to ice particles
- Strong winds – blow – up, down – enable rubbing – water drops / ice particles
- Learnt – chapter – exploring forces – 2 obj. – rub together – get charged
- This case – strong winds – blow – up, down – rub together – elec. charges develop – inside clouds
- Positive charged ice particles – lighter – move up – occupy upper part
- Negative charged water drops – heavier – move down – occupy lower part
- Charge separation – within clouds – ALSO – negative charged lower part – cloud – move closer – ground, nearby obj. – turn to positive charge
- Normally – air – elec. insulator – opp. charge – do not meet

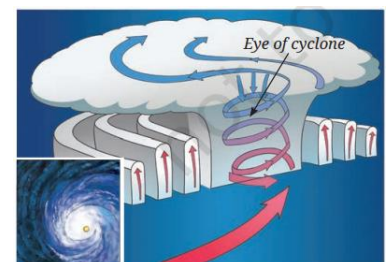
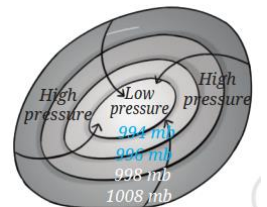


- BUT – build up – charges – very large – insulating prop. – air – break down
- Sudden flow – charges – produce – bright flash – light – **lightning**
- Lightning – opp. charge – collide – within clouds, b/w clouds and b/w clouds, grounds
- Lightning – heats air – cause air – expand – produce loud sound – **thunder**
- Storm with lightning, thunder – **thunderstorm**
- **A Step Further –**
 - Isolated, localized thunderstorms – sometimes – various regions – India
 - Thunderstorms – various names – *Kalboishakhi* – West Bengal, Bihar, Jharkhand – *Bordoisila* – Assam
 - Occurrence – before monsoon – help kharif crops
 - Kerala, Karnataka, Tamil Nadu – mango showers – help – ripening – mangoes
 - Karnataka – local thunderstorms – help growth – coffee plants
- Lightning – dangerous – ignite fires, damage buildings, cause severe burns – humans, animals
- Take precautions – protection – lightning – stay away – tall obj. – find – low-lying open area – crouch down – minimize contact – ground
- Avoid – laying flat – using umbrella – metallic rod – going in water – inside bus, car – much safer
- **Interesting fact –**
 - Lightning conductor – metallic rod – along walls – buildings
 - One end – pointed – higher – highest pt. – building
 - Other end – buried deep in ground – rod – easy path – transfer – elec. charge – ground



Cyclone

- **Cyclones** – large storms – over warm ocean waters
- Ocean water – heats up – warm, moist air – rise – water vapour – condense – form raindrops
- Evaporation – water – take heat – convert to vapour
- Water vapour – condense – raindrops – heat – release back – atm.
- Ascending air – warms further – rise further – even lower pressure
- Air – surrounding region – rush in – also rise
- Earth's rotation – cause – moving air – spin – this cycle – repeat – result – creation – very low-pressure area – high-sp. winds – revolve around
- Spinning sys. – clouds, winds, rain – **cyclone**
- Cyclone – region – lowest pressure – centre – **eye of cyclone**
- Eye of cyclone – wind – calm – surrounding region – strong winds, heavy rainfall
- Cyclone – ocean to land – generate higher wind sp. – compared – regular thunderstorms
- Cyclone – reach land – src. – moist air – cutoff – lose strength
- Cyclone – lose strength – travel over land – BUT – trail – destruction – month, years – repair
- Amphan cyclone – 2020 – peak wind sp. – 270 km/h
- Strong winds – cyclone – push ocean water to shore – create water wall – 3-12 m
- This surge – water – flood coastal areas – even far off areas
- Heavy rainfall – accompany cyclone – rivers overflow – trigger landslides
- Seawater – rush inland – contaminate drinking water src. – damage farmland



- Salt – seawater – soil less fertile – affect crops – roads blocked – fallen trees, debris – difficult to reach – affected areas
- Imp. – stay updated – weather reports – periodic alerts, warnings – IMD
- Thanks – weather satellites – track cyclones – predict path – help reduce impact – life, prop.
- Live – cyclone-prone area – keep emergency kit ready – essential items – during cyclone – move – designated shelters